PRBS Test via Loopback mode on PHY Application Note

1. Introduction

The Diodes' PCIe GEN3 Packet Switch's PHY has embedded BERT modules, which transmit PRBS patterns out from PHY, and receive PRBS streams and check the correctness. The modules are used to inspect the signal integrity of the link. In the process of setting up the link and testing the modules, a loopback test fixture or the link partner's PHY in remote loopback mode is required for onboard external loopback, while a Tx side 49.9Ω termination is required for onboard internal loopback.

Each switch tile within the Packet Switch contains 4 Quad PHY's, labeled as PHY0, PHY1, PHY2 and PHY3 respectively in this document. In addition, each Quad PHY's macro is comprised of 4 lanes (i.e. Lane 0 to 3). Each lane can be tested with external or internal PRBS patterns independently.

2. Procedures to Test External Loopback PRBS via I2C Commands

2.1. Following is an example for enabling PRBS test on Lane N of any Quad PHY using a Loopback Test Fixture

Select Port for settings

STEP	CMD	OFFSET	VALUE	COMMENT
1.	CFGW	390[Note1]	x	x is bit[2] & set "1" : Force LTSSM state stay in detect state (force2detect)
2.	Wait 1 ms			
3.	CFGW	98[Note2]	xxxx	Set bit[3:0] : 0011b is gen3, 0010b is gen2, 0001b is gen1 (Select link speed)
4.	CFGW	98[Note2]	1xxxx	Set bit[4]=1: Enter compliance
5.	CFGW	390[Note1]	x	x is bit[2] & set "0" : Force LTSSM state stay leave detect state

Select PHY and Lane for settings

STEP	CMD	OFFSET	VALUE	COMMENT
6.	CR_W	1N02 Note3	xxxx	Preset_setting for TX EQ (see Table 1: Preset_setting) (CR_R/CR_W, see
7.	CR_W	1N03 Note3	xxxx	Access PHY Registers via CR Interface)
8.	CR_W	1N40	0CFE	RX power state control for P0
9.	CR_W	1N28[Note4]	yyyx	x=1 to b, set PHY TX BERT GENERATOR (see Table 2: PRBS Bert Setting) yyy=000 when x=1 to 8, yyy=pattern when x=9 to b (see Table 3: Pattern)
10.	CR_W	3N08	1000	Enable RX EQ Adaption Request (override the value from PCS)
11.	CR_W	3N08	1800	Set RX EQ Adaption Request
12.	Wait 12 ms			
13.	CR_R	3N10		bit[0]=1: RX EQ Adaption is Acked (CCA)
14.	Wait 20 ms			
15.	CR_W	3N10	0000	clear bit[0]
16.	CR_W	1N51	000x	x=1 to b, Set PHY RX BERT MATCHER (see Table 2: PRBS Bert Setting)
17.	CR_W	1N51	001x	x=1 to b, Synchronize pattern matcher PRBS with incoming data (see Table 2: PRBS Bert Setting)
18.	CR_R	1N52		Read out ERR_BIT_CNT expressed by bit [14:0]

Note: x value in step 9, 15 and 16 should be the same.
N value in step 6~11 and 13~17 should be the same, and indicates the Lane# (0~3).

2.2. Following is an example for enabling PRBS test on Lane N of any Quad PHY and link partner's PHY in remote loopback mode (Root Complex ←→ DIODE-A ←→ DIODE-B)

1st : DIODE-A

Select Downstream Port for settings

STEP	CMD	OFFSET	VALUE	COMMENT
1.	CFGW	390[Note1]	x	x is bit[2] & set "1" : Force LTSSM state stay in detect state (force2detect)
2.	CFGW	3BC[Note1]	x	x is bit[3] & set "1" :: Used to control poll.compliance exit
3.	CFGW	98[Note2]	xxxx	Set bit[3:0] : 0011b is gen3, 0010b is gen2, 0001b is gen1 (Select link speed)
4.	CFGW	390[Note1]	x	x is bit[2] & set "0" : Force LTSSM state stay leave detect state

Select Upstream Port for settings

STEP	CMD	OFFSET	VALUE	COMMENT
1.	CFGW	310	2x0	Bit[4:0] --> Debug mode select, 0 --> LTSSM Bit[8:5] --> Debug port select to port x (x = Diodes-A Downstream Port) Bit[9] --> Start to select debug output data
2.	CFGR	314[Note5]	02	Check Value = 0x2h (Polling state)

2nd : DIODE-B (Remote Loopback mode)

Select Link Port for settings

STEP	CMD	OFFSET	VALUE	COMMENT
1.	CFGW	390[Note1]	x	x is bit[2] & set "1" : Force LTSSM state stay in detect state (force2detect)
2.	CFGW	3BC[Note1]	x	x is bit[3] & set "1" :: Used to control poll.compliance exit
3.	CFGW	98[Note2]	xxxx	Set bit[3:0] : 0011b is gen3, 0010b is gen2, 0001b is gen1 (Select link speed)
4.	CFGW	390[Note1]	x	x is bit[2] & set "0" : Force LTSSM state stay leave detect state

Select Upstream Port for settings

STEP	CMD	OFFSET	VALUE	COMMENT
1.	CFGW	310	2x0	Bit[4:0] --> Debug mode select, 0 --> LTSSM Bit[8:5] --> Debug port select to port x (x = Diodes-A Downstream Port) Bit[9] --> Start to select debug output data
2.	CFGR	314[Note5]	02	Check Value = 0x2h (Polling state)

Select PHY and Lane for settings

STEP	CMD	OFFSET	VALUE	COMMENT
5.	CR_W	1N02[Note3]	xxxx	Preset_setting for TX EQ (see Table 1: Preset_setting) (CR_R/CR_W, see Access PHY Registers via CR Interface)
6.	CR_W	1N03 [Note3]	xxxx	Preset_setting for TX EQ(see Table 1: Preset_setting)
7.	CR_W	1N05	4004	Enable Rx Data Path, N = Lane#
8.	CR_W	1N00	0006	Enable Rx-to-Tx loopback path, N = Lane#

Note: N value in step 5~8 should be the same, and indicates the Lane# (0~3).

3rd : DIODE-A

Select PHY and Lane for settings

STEP	CMD	OFFSET	VALUE	COMMENT
1.	CR_W	1N02[Note3]	xxxx	Preset_setting for TX EQ (see Table 1: Preset_setting) (CR_R/CR_W, see Access PHY Registers via CR Interface)
2.	CR_W	1N03 [Note3]	xxxx	Preset_setting for TX EQ(see Table 1: Preset_setting)
3.	CR_W	1N40	0CFE	RX power state control for P0
4.	CR_W	1N28[Note4]	yyyx	x=1 to b, set PHY TX BERT GENERATOR (see Table 2: PRBS Bert Setting) yyy=000 when x=1 to 8, yyy=pattern when x=9 to b (see Table 3: Pattern)
5.	CR_W	1N51	000x	x=1 to b, Set PHY RX BERT MATCHER (see Table 2: PRBS Bert Setting)
6.	CR_W	1N51	001x	x=1 to b, Synchronize pattern matcher PRBS with incoming data (see Table 2: PRBS Bert Setting)
7	CR_R	1N52		Read out ERR_BIT_CNT expressed by bit [14:0]

Note: x value in step 4~6 should be the same.
N value in step 1~7 should be the same, and indicates the Lane# (0~3).

Note1. CFG register at offset 390h and 3BCh is LTSSM control register. The purpose of the setting is to ensure that the connection is maintained at compliance mode.

Note2. CFG register at offset 98h is link control 2 register. By writing 1x to bit [4:0], it can decide the link speed for testing PRBS or other data patterns. As the CFG register is defined in PCIe port, please refer to Table 5-4 port-lane mapping in datasheet to select the corresponding port for accessing its link control 2 register. Also, the normalized lane number in each Quad PHY is re-mapped from the lane number of Table 4-1 as below (see The Mapping between Lane Number and PHY Number/N Value).

Note3. Please refer to The Mapping between Lane Number and PHY Number/N Value

Note4. Please refer to PHY Registers used for BERT

Note5. When the value of 02h is not read, please set 310h again, and then read 314 again. When the value of 02h has not read more than three times, restart packet switch and re-run PRBS procedure.

3. Procedures to Test Internal Loopback PRBS via I2C Commands

3.1. Following is an example for enabling Internal Loopback PRBS test on Lane N of any Quad PHY (Tx side requires a loading (to be connected to an EP or Root Complex) to eliminate the external I/O pad interference during the test. If there is no loading, adding a 49.9Ω termination is suggested)

Select Port for settings

STEP	CMD	OFFSET	VALUE	COMMENT
1.	CFGW	390[Note1]	x	x is bit[2] & set "1" : Force LTSSM state stay in detect state (force2detect)
2.	CFGW	3BC[Note1]	x	x is bit[3] & set "1" :: Used to control poll.compliance exit
3.	CFGW	98[Note2]	xxxx	Set bit[3:0] : 0011b is gen3, 0010b is gen2, 0001b is gen1 (Select link speed)
4.	CFGW	390[Note1]	x	x is bit[2] & set "0" : Force LTSSM state stay leave detect state

Select PHY and Lane for settings

STEP	CMD	OFFSET	VALUE	COMMENT
5.	CR_W	1N00	0005	Enable Internal TX to RX Serial Loopback path
6.	CR_W	1N28[Note4]	yyyx	x=1 to b, set PHY TX BERT GENERATOR (see Table 2: PRBS Bert Setting) yyy=000 when x=1 to 8, yyy=pattern when x=9 to b (see Table 3: Pattern)
7.	CR_W	1N51	000x	x=1 to b, Set PHY RX BERT MATCHER (see Table 2: PRBS Bert Setting)
8.	CR_W	1N51	001x	x=1 to b, Synchronize pattern matcher PRBS with incoming data (see Table 2: PRBS Bert Setting)
9.	CR_R	1N52		Read out ERR_BIT_CNT expressed by bit [14:0]

Note: x value in step 6, 7 and 8 should be the same.
N value in step 5~9 should be the same, and indicates the Lane# (0~3).

Note1. CFG register at offset 390h and 3BCh is LTSSM control register. The purpose of the setting is to ensure that the connection is maintained at compliance mode.

Note2. CFG register at offset 98h is link control 2 register. By writing 1x to bit [4:0], it can decide the link speed for testing PRBS or other data patterns. As the CFG register is defined in PCIe port, please refer to Table 5-4 port-lane mapping in datasheet to select the corresponding port for accessing its link control 2 register. Also, the normalized lane number in each Quad PHY is re-mapped from the lane number of Table 4-1 as below (see The Mapping between Lane Number and PHY Number/N Value).

Note3. Please refer to The Mapping between Lane Number and PHY Number/N Value

Note4. Please refer to PHY Registers used for BERT

4. Access PHY Registers via CR Interface

CR_R and CR_W are an indirect access method from CFG registers to internal PHY registers through CR interface. The CFG registers located at offset from 5C0h to 5D0h are the windows for accessing PHY registers whose address format is defined in the section of Address format for CR_W and CR_R.

4.1. Definitions for CFG Offset 5C0h to 5D0h

.PHYN CONTROL REGISTER: OFFSET: 0x05C0

	31:20	19	18	17	16	15:12	11
	Reserved	PHY3_WR_STS	PHY2_WR_STS	PHY1_WR_STS	PHY0_WR_STS	Reserved	PHY3_R_EN
10	9	8	7:4	3	2	1	0
PHY2_R_EN	PHY1_R_EN	PHY0_R_EN	Reserved	PHY3_W_EN	PHY2_W_EN	PHY1_W_EN	PHY0_W_EN

.PHY0 ADDRESS and DATA REGISTER: OFFSET: 0x05C4

31:16	15:0
PHY_REG_ADDRESS	PHY_RW_DATA

.PHY1 ADDRESS and DATA REGISTER: OFFSET: 0x05C8

31:16	15:0
PHY_REG_ADDRESS	PHY_RW_DATA

.PHY2 ADDRESS and DATA REGISTER: OFFSET: 0x05CC

31:16	15:0
PHY_REG_ADDRESS	PHY_RW_DATA

.PHY3 ADDRESS and DATA REGISTER: OFFSET: 0x05D0

31:16	15:0
PHY_REG_ADDRESS	PHY_RW_DATA

4.2. The Mapping between Lane Number and PHY Number/N Value

The PHY Control Registers of any one can be accessed through the CR CTRL register defined in the table below. A specific lane can be located by enumerating the PHY Number from 0 to 3, and the N Value from 0 to 3.

Table 4-1 PCIe Configuration Registers Used to Access PHY CR Registers

Lane	PHY Number	N Value	PCIe Configuration Register Used to Access PHY CR Register	3G1632GP	3G816GP	3G808GP	3G606GP
Lane 0	0	0	Offset 5C0h and 5C4h registers	0	0	0	0
Lane 1	0	1	Offset 5C0h and 5C4h registers	0	0	0	0
Lane 2	0	2	Offset 5C0h and 5C4h registers	0	0	0	
Lane 3	0	3	Offset 5C0h and 5C4h registers	0	0	0	
Lane 4	1	0	Offset 5C0h and 5C8h registers	1	1		
Lane 5	1	1	Offset 5C0h and 5C8h registers	1	1		
Lane 6	1	2	Offset 5C0h and 5C8h registers	1	1		
Lane 7	1	3	Offset 5C0h and 5C8h registers	1	1		
Lane 8	2	0	Offset 5C0h and 5CCh registers	2	2	2	2
Lane 9	2	1	Offset 5C0h and 5CCh registers	2	2	2	2
Lane 10	2	2	Offset 5C0h and 5CCh registers	2	2	2	2
Lane 11	2	3	Offset 5C0h and 5CCh registers	2	2	2	2
Lane 12	3	0	Offset 5C0h and 5D0h registers	3	3		
Lane 13	3	1	Offset 5C0h and 5D0h registers	3	3		
Lane 14	3	2	Offset 5C0h and 5D0h registers	3	3		
Lane 15	3	3	Offset 5C0h and 5D0h registers	3	3		

The packet switch provides CFG registers (offset 5C0h~5D0h) to access Quad PHY registers. Those registers are supported for P0 only.

4.3. CR_R Procedures (read PHY register by CR interface)

(1) Check PHY0~PHY3 CR Status by reading CFG Offset 5C0h
Bit [19] =1, PHY3 CR READY for access
Bit [18] =1, PHY2 CR READY for access
Bit [17] =1, PHY1 CR READY for access
Bit [16] =1, PHY0 CR READY for access

(2) Write PHY0~PHY3 CR Address by CFG OFFSET 5C4h, 5C8h, 5CCh and 5D0h
Bit [31:16], PHY Register Address

(3) Write PHY0 ~PHY3 CR Read Enable bit by CFG OFFSET 5C0h
Bit [11] =1, Enable PHY3 CR Read
Bit [10] =1, Enable PHY2 CR Read
Bit [9] =1, Enable PHY1 CR Read
Bit [8] =1, Enable PHY0 CR Read

(4) Read PHY0 ~PHY3 CR Read Data by CFG OFFSET 5C4h, 5C8h, 5CCh and 5D0h
Bit [15:0], PHY Register Data

4.4. CR_W Procedures (write PHY register by CR interface)

(1) Check PHY0 ~PHY3 CR STATUS by reading CFG OFFSET 5C0h
If Bit [19] =1, PHY3 CR READY for access
If Bit [18] =1, PHY2 CR READY for access
If Bit [17] =1, PHY1 CR READY for access
If Bit [16] =1, PHY0 CR READY for access

(2) Write PHY0~PHY3 CR Address and Data by CFG OFFSET 5C4h, 5C8h, 5CCh and 5D0h
Bit [31:16], PHY Register Address
Bit [15:0], PHY Register Data

(3) Enable PHY0 ~PHY3 CR Write Enable bit by CFG OFFSET 5C0h
Bit [3] =1, Enable PHY3 CR Write
Bit [2] =1, Enable PHY2 CR Write
Bit [1] =1, Enable PHY1 CR Write
Bit [0] =1, Enable PHY0 CR Write

4.5. CR_W Example

CR_W 1002 b9f8 for Phy0, Lane0
CFGW 0x5c4 32'h1002b9f8
CFGW 0x5c0 32'h00000001

4.6. CR_R Example

CR_R 1052 for Phy0, Lane0
CFGW 0x5c4 32'h10520000
CFGW 0x5c0 32'h00000100
CFGR 0x5c4 32'h1052XXXX (XXXX is read value)

5. Tables

5.1. Table 1: Preset_setting

GEN3	1002	1003
Preset 0	BDF8	3440
Preset 1	C3F8	2E40
Preset 2	C1F8	3040
Preset 3	C7F8	2A40
Preset 4	D1F8	2040
Preset 5	C9F8	2050
Preset 6	C7F8	2054
Preset 7	B9F8	3050
Preset 8	BDF8	2A54
Preset 9	C3F8	205C
Preset 10	B7F8	3A40

GEN2	1002	1003	Tx De-emphasis
	BDF8	3440	-6db
	C3F8	2E40	-3.5db

GEN1	1002	1003	Tx De-emphasis
	C3F8	2E40	-3.5db

5.2. Table 2: PRBS Bert Setting

Value	Formula
1	PRBS31: X^31+X^28+1
2	PRBS23: X^23+X^18+1
3	PRBS23: X^23+X^21+X^16+X^8+X^5+X^2+1
4	PRBS16: X^16+X^5+X^4+X^3+1
5	PRBS15: X^15+X^14+1
6	PRBS11: X^11+X^9+1
7	PRBS9: X^9+X^5+1
8	PRBS7: X^7+X^6+1
9	Fixed word (PAT0) (see below table 3)
a	DC balance word (PAT0, ~PAT0)
b	Fixed pattern (000, PAT0, 3fff, ~PAT0)

5.3. Table 3: Pattern

Bit[14:5] is used to set PAT0 (pattern).

6. PHY Registers used for BERT

0x1N28: TX BERT Pattern Generation Control for Lane N

Bits	Name	Type	Description	Default
15	Reserved	RO	Reserved for future use	0
14:5	PAT0	R/W	Pattern for modes 9-11	0
4	Trigger Error	R/W	Insert a single error into a LSB Any write of a "1" to this bit will insert an error.	0
3:0	Mode	R/W	Pattern for generation When changing modes, it must be changed to disabled at first. ■ 0 – Disabled ■ 1 – LFSR31: X^31 + X^28 + 1 ■ 2 – LFSR23: X^23 + X^18 + 1 ■ 3 – LFSR23: X^23 + X^21 +X^16 + X^8 + X^5 + X^2 +1 ■ 4 – LFSR16: X^16 + X^5 +X^4 + X^3 +1 ■ 5 – LFSR15: X^15 + X^14 + 1 ■ 6 – LFSR11: X^11 + X^9 + 1 ■ 7 – LFSR9: X^9 + X^5 + 1 ■ 8 – LFSR7: X^7 + X^6 + 1 ■ 9 – Fixed word (PAT0) ■ 10 – DC balanced word (PAT0, ~PAT0) ■ 11 – Fixed pattern (000, PAT0, 3ff, ~PAT0)	0

0x1N51: RX BERT Pattern Matcher Control for Lane N

Bits	Name	Type	Description	Default
15:5	Reserved	RO	Reserved for future use	0
4	SYNC	R/W	Synchronize pattern matcher LFSR with incoming data. A write of a one to this bit will reset the error counter and start a synchronization of the pattern matcher. There is no need to write this back to zero to run normally.	0
3:0	Mode	R/W	Pattern for matching When changing modes, it must be changed to disabled at first. ■ 0 – Disabled ■ 1 – LFSR31: X^31 + X^28 + 1 ■ 2 – LFSR23: X^23 + X^18 + 1 ■ 3 – LFSR23: X^23 + X^21 +X^16 + X^8 + X^5 + X^2 +1 ■ 4 – LFSR16: X^16 + X^5 +X^4 + X^3 +1 ■ 5 – LFSR15: X^15 + X^14 + 1 ■ 6 – LFSR11: X^11 + X^9 + 1 ■ 7 – LFSR9: X^9 + X^5 + 1 ■ 8 – LFSR7: X^7 + X^6 + 1 ■ 9 – D[n] = D[n-10] ■ 10 – D[n] = !D[n-10] ■ 11 – D[n] = !D[n-20]	0

0x1N52: RX BERT Pattern Match Error Counter for Lane N

Bits	Name	Type	Description	Default
15	OV14	R/W	If active, multiply COUNT by 128 If OV14= 1 and COUNT= 2^15 - 1, signals overflow of counter(volatile and 2 reads needed to read value)	0
14:0	COUNT	R/W	A read of this register, or sync of the pattern matcher resets the error counter. Current error count: If OV14 field is active, then multiply count by 128(volatile and 2 reads needed to read value)	0

7. Grade Debug out register setup

9.3.125 DEBUGOUT CONTROL REGISTER – OFFSET 310h (Port 0 Only)

BIT	FUNCTION	TYPE	DESCRIPTION	EEPROM/ I2C-SMBUS	DEFAULT
4:0	Debug Mode Select	RW	Debug mode select. Selects a signal group for probing the current internal status. For example, "0" represents LTSSM signal group. As to other values, please inquire internal team for further information.	Yes	0_0000b
7:5	Debug Port_Select_S1	RW	Debug port select s1. Selects a port number for monitoring at a given signal group.	Yes	000b
8	DebugPort_ Select_S2	RW	Debugport select s2.	Yes	0
9	Debug Output Start	RW	Start to capture debug output data.	Yes	0
31:10	Reserved	RsvdP	Not support.	No	0-0h

9.3.126 DEBUGOUT DATA REGISTER – OFFSET 314h (Port 0 Only)

BIT	FUNCTION	TYPE	DESCRIPTION	EEPROM/ I2C-SMBUS	DEFAULT
9:0	Debug Output Data	RO	Content of the debug output data. For example, if LTSSM signal group is selected, the meaning of debug output data is as follows. 001h: detect 002h: polling 004h: configuration 008h: L0 010h: L1 020h: L2 040h: disable 080h: hot-reset 100h: loopback 200h: recovery Others: Reserved	No	000h
31:10	Reserved	RsvdP	Not support	No	0000_0h

Example 1, Port 0 LTSSM check
All register write/read under port0
1. Write Port 0 offset 0x310h as 0x200 // Bit[4:0] --> Debug mode select, 0 --> LTSSM
// Bit[8:5] --> Debug port select to port 0
// Bit[9] --> Start to select debug output data
2. Read Port 0 offset 0x314Ch // Value 0x8h which L0 state under link is up with RC

Example 2, Port 6 LTSSM check
All register write/read under port0
1. Write Port 0 offset 0x310h as 0x2C0 // Bit[4:0] --> Debug mode select, 0 --> LTSSM
// Bit[8:5] --> Debug port select to port 6
// Bit[9] --> Start to select debug output data
2. Read Port 0 offset 0x314Ch

Revision History

Date	Revision	Description
2021/11/24	0.1f	Preliminary Application Note
2022/10/21	0.1g	Section 2.2 Remove 3rd : DIODE-A STEP about 3N08 & 3N10
2022/11/09	0.2	Add Section 3 Procedures to Test Internal Loopback PRBS via I2C Commands Change file name to “PRBS Test via Loopback mode Application Note”
2023/06/14	0.2a	Add a step to read the LTSSM status to ensure that the Packet Switch is currently linked in the polling state
2024/11/15	0.2b	Modify 98h/390h/3bch numerical representation